IoT Challenge #3 - Node-RED Report

Team: YOUR NAME SURNAME + TEAMMATE NAME SURNAME
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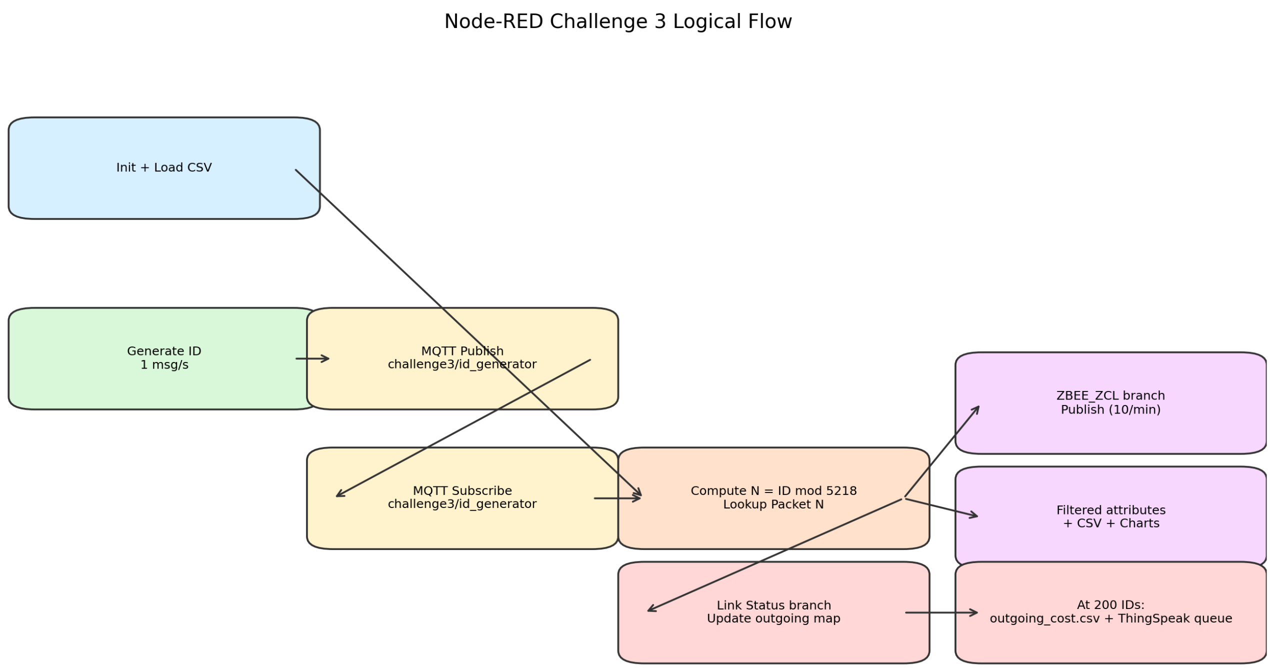
1) Implementation overview

The Node-RED flow publishes one random ID per second to the local broker (localhost:1884, topic challenge3/id_generator), subscribes to the same topic, computes N = ID mod 5218, and processes packet N from challenge3.csv. ZBEE_ZCL packets are republished with the required payload and rate-limited at 10 messages/minute; RMS/Active Power attributes are extracted to filtered_elems.csv and plotted on dashboard charts. Link Status packets update outgoing link costs; at exactly 200 received IDs the flow stops, writes outgoing_cost.csv, and builds the ThingSpeak queue ordered by destination address.

2) Node-RED flow image



3) Main node blocks and meaning

A. Init + Load CSV: resets counters/context and writes CSV headers.

B. Generator branch: creates random IDs [0..30000], JSON payload, and id_log rows.

C. Subscription branch: counts incoming IDs (hard stop at 200) and computes modulo index N.

D. ZBEE_ZCL branch: builds required publish payload and enforces 10/min rate limit.

E. Filter branch: extracts Active Power / RMS Current / RMS Voltage with positional matching and logs CSV rows.

F. Link Status branch: updates per-source per-destination outgoing cost map.

G. Finalization branch: writes outgoing_cost.csv and queues ThingSpeak HTTP updates every 20s.

4) Generated output summary (200-message run)

Metric	Value
Seed	20260430
Start timestamp	1777559569
Messages processed	200
ZBEE_ZCL packets	120
Link Status packets	55
Ignored packets	25
filtered_elems rows	26
outgoing_cost rows	70
ThingSpeak queue rows	8
Smallest source (ThingSpeak)	0x0000

5) CSV excerpts

id_log.csv (first rows)

No.	ID	TIMESTAMP
1	7158	1777902320
2	646	1777902321
3	14368	1777902322
4	17263	1777902323
5	21338	1777902324
6	28086	1777902325
7	3241	1777902326
8	15837	1777902327

filtered_elems.csv (first rows)

No.	Timestamp	Sequence Number	Attribute	Status	Data Type	Data Value
1	1777902329	92	Active Power	Success	16-Bit Signed Integer	0
2	1777902329	92	RMS Current	Success	16-Bit Unsigned Integer	0
3	1777902329	92	RMS Voltage	Success	16-Bit Unsigned Integer	221
4	1777902338	139	Active Power	Success	16-Bit Signed Integer	0
5	1777902338	139	RMS Current	Success	16-Bit Unsigned Integer	0
6	1777902338	139	RMS Voltage	Success	16-Bit Unsigned Integer	221
7	1777902342	164	Active Power	Success	16-Bit Signed Integer	0
8	1777902342	164	RMS Current	Success	16-Bit Unsigned Integer	0

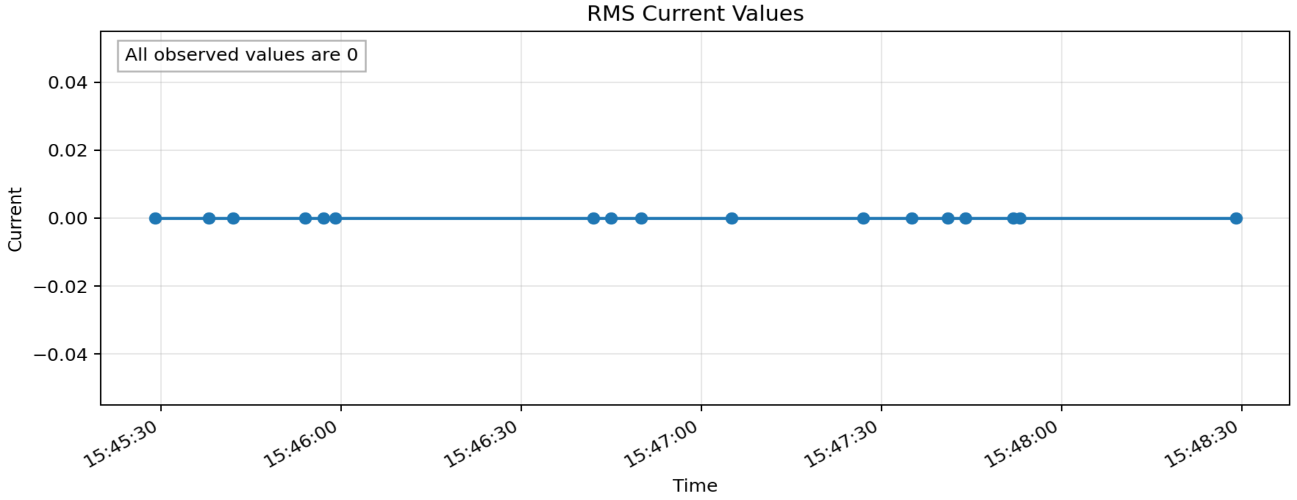
No.	Source	Destination	Cost
1	0x0000	0x0112	3
2	0x0000	0x1cd8	1
3	0x0000	0x1e15	3

4	0x0000	0x3d95	3
5	0x0000	0x4e11	3
6	0x0000	0xa706	5
7	0x0000	0xe1a6	1
8	0x0000	0xec7f	1

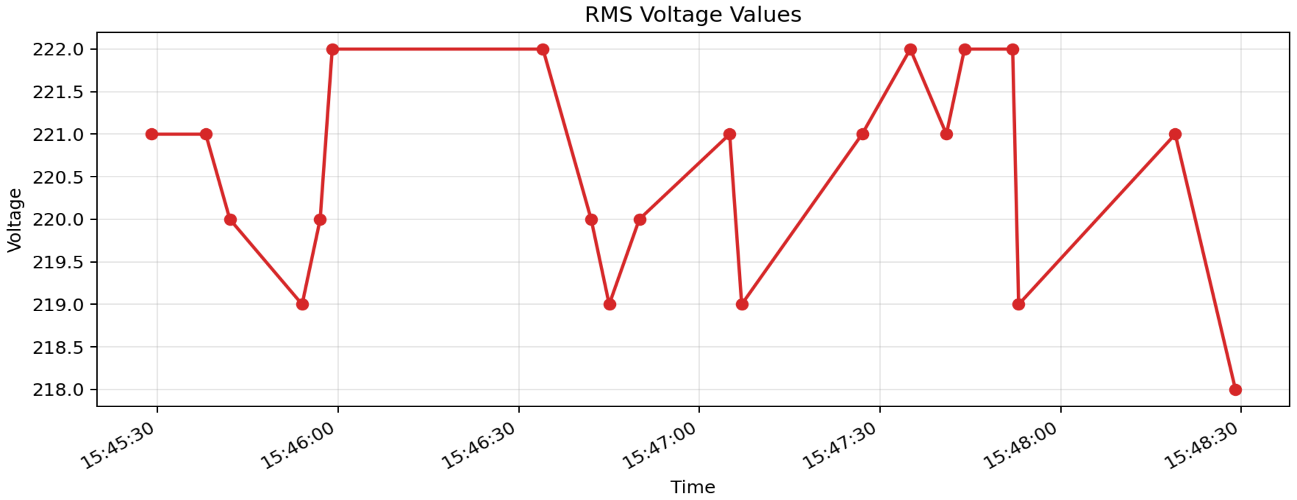
6) RMS chart outputs

RMS Current samples: 17, unique values: [0]. RMS Voltage samples: 20, unique values: [218, 219, 220, 221, 222].

RMS Current chart



RMS Voltage chart



7) ThingSpeak channel note

Flow includes ThingSpeak HTTP updates every 20 seconds in destination-address order. Set flow.thingspeak_write_api_key (or THINGSPEAK_WRITE_API_KEY) and add the public channel URL here before final submission: **TO_BE_FILLED_WITH_PUBLIC_CHANNEL_LINK**.